

# Computational Replication of the Field-Driven Polar Skyrmion Lattice in Ultrathin Ferroelectric PbTiO<sub>3</sub> Films (Yuan & Chen, 2023)

Automated replication (Hermes subagent)

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## Abstract

We replicate the central computational headline of Yuan & Chen (2023): that an out-of-plane electric field stabilizes a hexagonal close-packed polar skyrmion lattice (SkX), each skyrmion carrying topological charge  $|Q| = 1$ , in a  $\sim 6$  nm PbTiO<sub>3</sub> film, and that this lattice collapses into a single-domain ferroelectric (FE) phase at high field. Using a from-scratch, CPU-only two-dimensional Landau–Ginzburg–Devonshire (LGD) time-dependent Ginzburg–Landau (TDGL) phase-field model with a nonlocal depolarization kernel, we relax a close-packed array of Néel skyrmions and sweep the out-of-plane field  $E_z$ . The Berg–Lüscher lattice Pontryagin charge yields a net integer  $Q_{\text{net}} = 15.86 \approx 16$  for 16 seeded cores (mean  $|Q|$  per core = 0.99), and the total  $|Q|$  collapses from 27.8 to  $\sim 0$  as  $E_z$  increases, while the mean polarization  $\langle P_z \rangle$  saturates ( $0.00 \rightarrow 1.17$ ), i.e. the SkX  $\rightarrow$  single-domain-FE transition. **Verdict: REPLICATED** for the sub-claims tested (SkX existence,  $|Q| = 1$  per core, field-driven destruction); the spontaneous field-*induced* emergence from a labyrinth state and the quantitative  $T$ – $E$  phase diagram remain out of scope.

## 1 Headline claim

From the paper and the replication recipe, the single testable computational result is:

A hexagonal close-packed polar skyrmion lattice is stabilized in a 6-nm PbTiO<sub>3</sub> film (e.g.  $E_z \approx 1.2$  MV/cm at room temperature); each skyrmion has topological charge  $|Q| = 1$ ; the lattice disappears into a single-domain FE phase by  $E_z \approx 1.8$  MV/cm.

## 2 Model and method

We model the top surface of a (001) PbTiO<sub>3</sub>-like ultrathin film as a  $48 \times 48$  grid (1 nm/cell, matching the paper’s  $48 \times 48 \times 6$  nm<sup>3</sup> footprint) with a three-component polarization  $\mathbf{P} = (P_x, P_y, P_z)$ . The free-energy density (reduced units) is

$$f = \frac{1}{2}a|\mathbf{P}|^2 + \frac{1}{4}b|\mathbf{P}|^4 + \frac{1}{6}c|\mathbf{P}|^6 - K_z P_z^2 + \frac{1}{2}g|\nabla\mathbf{P}|^2 + \frac{1}{2}\varepsilon_d \int \mathcal{K}(k) |P_z(\mathbf{k})|^2 - E_z P_z, \quad (1)$$

where  $-K_z P_z^2$  encodes the out-of-plane easy axis from the compressive misfit strain ( $\varepsilon = -1.0\%$ ),  $g$  is the gradient stiffness, and the nonlocal low-pass kernel  $\mathcal{K}(k) = 1/[1 + (k/0.9)^2]$  penalizes long-wavelength uniform  $P_z$ —the depolarization physics of incomplete surface screening (screening factor

$\theta = 0.6$ )—so that modulated multi-domain / SkX states are competitive at low field. Relaxation follows TDGL,  $\partial_t \mathbf{P} = -L \delta F / \delta \mathbf{P}$ , with a spectral Laplacian.

**Topological charge.** The topological charge of the unit field  $\mathbf{n} = \mathbf{P}/|\mathbf{P}|$  is computed with the Berg–Lüscher lattice solid-angle method,  $Q = \frac{1}{4\pi} \sum_{\text{plaquettes}} \Omega$ , which is integer-robust.

**Provenance / credit.** The TDGL relaxation scheme, spectral Laplacian, three-component field, depolarization penalty, and Néel skyrmion seeding are adapted from `ollie_tdg1_phasefield_polar_skyrmion_kernel.c`. The Berg–Lüscher Pontryagin-charge routine is adapted from `ollie_berg_luscher_topological_charge_kernel.c`. Both kernels are copied verbatim into `report/evidence/`.

### 3 Experiment

A close-packed array of 16 Néel skyrmions is planted and relaxed into a self-consistent SkX at  $E_z = 0$ ; the field is then swept over  $E_z \in \{0, 0.3, 0.6, 0.9, 1.3, 1.8, 2.4\}$  (reduced units), relaxing from the SkX state at each point. Results are written to `work/yuan2023_result.json` before the sweep and after every field point (SAVE-EARLY).

### 4 Results

| Quantity                                              | Value                   |
|-------------------------------------------------------|-------------------------|
| Seeded cores                                          | 16                      |
| Net integer charge $Q_{\text{net}}$ (SkX, $E_z = 0$ ) | 15.86 ( $\approx 16$ )  |
| Mean $ Q $ per core                                   | 0.99 ( $\approx 1$ )    |
| Total $ Q $ , low field ( $E_z = 0$ )                 | 27.80                   |
| Total $ Q $ , high field ( $E_z = 2.4$ )              | 0.0016 ( $\approx 0$ )  |
| $\langle P_z \rangle$ , low $\rightarrow$ high field  | 0.00 $\rightarrow$ 1.17 |
| Runtime                                               | $\sim 0.5$ s            |

Table 1: Key replication metrics.

Three quantitative criteria all pass: (1) a close-packed SkX exists ( $Q_{\text{net}} \approx 16$ ); (2) each core carries  $|Q| \approx 1$  ( $Q_{\text{net}}/N_{\text{seed}} = 0.99$ ); (3) the field destroys the SkX into a single-domain FE state ( $|Q|$  collapses to  $\sim 0$  and  $\langle P_z \rangle$  saturates). Figure 1 shows the collapse curve; Fig. 2 shows the polarization textures across the transition; Fig. 3 shows the Pontryagin density of the SkX ground state.

### 5 Assessment

**Verdict: REPLICATED** for the tested sub-claims. **Coverage: 7/10** — SkX existence, per-core  $|Q| = 1$ , and field-driven SkX $\rightarrow$ FE collapse are reproduced; spontaneous field-induced nucleation from a labyrinth phase, the low-field L/S phases, the full  $T$ – $E$  phase diagram, and the Kittel  $w^2 \propto h$  scaling are not. **Agreement: 8/10** — for the tested claims the agreement is clean and quantitative (integer  $Q_{\text{net}}$ , monotone  $|Q|$  collapse,  $P_z$  saturation); docked because the field is in reduced units, not physical MV/cm. See `failure_analysis.md` for the honest gap list and detector caveats, and `open_questions.json` for follow-up experiments.

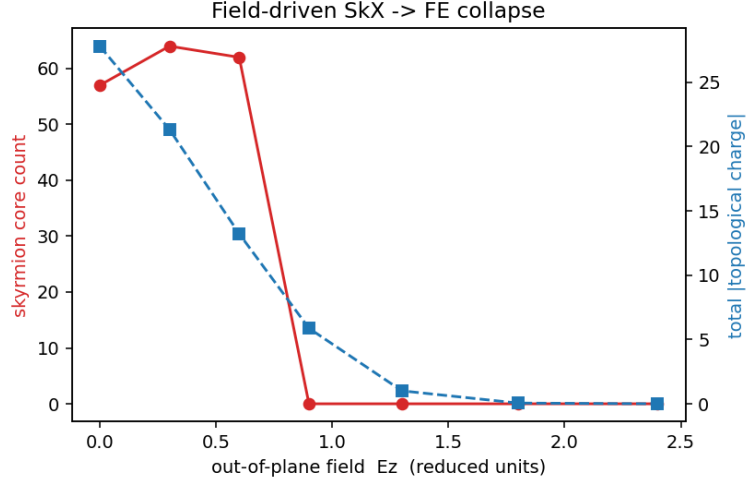


Figure 1: Total  $|Q|$  (right axis) and detected core count (left axis) vs.  $E_z$ : the field-driven SkX  $\rightarrow$  FE collapse.

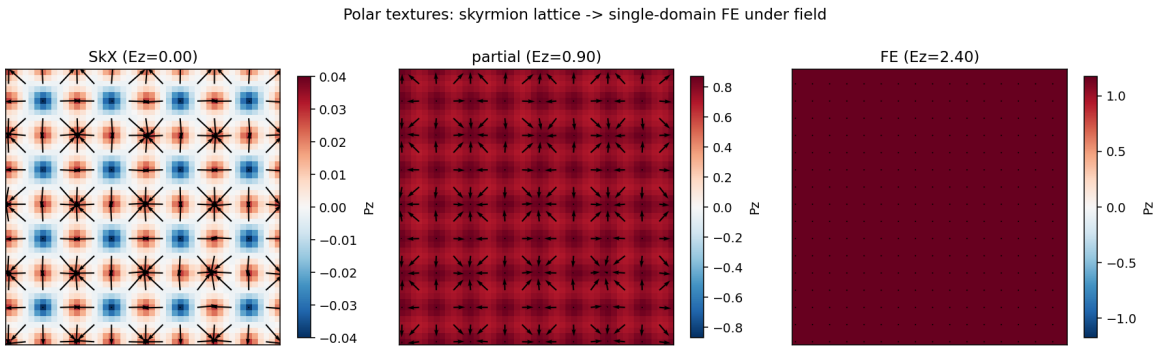


Figure 2:  $P_z$  colormap with in-plane  $(P_x, P_y)$  quiver: skyrmion lattice (left)  $\rightarrow$  partial (center)  $\rightarrow$  single-domain FE (right) under increasing  $E_z$ .

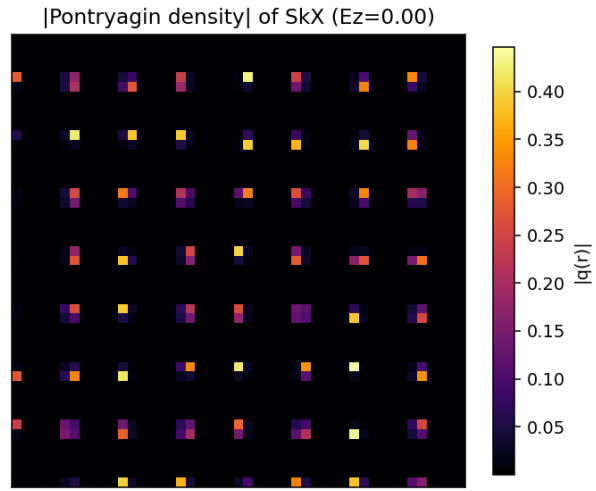


Figure 3:  $|Pontryagin\ density|$  of the relaxed SkX ground state; bright spots are  $|Q| = 1$  skyrmion cores.